



university of
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Models and Semantics of Computation (MaSC)

Lecture 5: Rule Induction

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Well-Founded Induction



Let $\prec \subseteq A \times A$ be a well-founded relation.

$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

- ▶ A general **proof principle**, aka Noetherian induction.
- ▶ Derived from Theorem 4.5, direction **(2)** $\Rightarrow$ **(1)**.



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- ▶ A **base case** is any element of A such that the set $\{b \in A \mid b \prec a\}$ is empty.



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We can **instantiate the principle**, by choosing specific A and $\prec$.

An Instance: Structural Induction



- Set: $A = T_\Sigma$ (closed terms)
- Well-founded relation: **immediate subterm relation**
 $\prec = \{(t_i, f(t_1, \dots, t_n)) \mid f \in \Sigma_n, i \in [1..n]\}$

$$\frac{\forall a \in A. ((\forall b \prec a. P(b)) \Rightarrow P(a))}{\forall a \in A. P(a)}$$

$\rightsquigarrow$

$$\frac{\forall n \in \mathbb{N}. \forall f \in \Sigma_n. \forall t_1, \dots, t_n. (P(t_1) \wedge \dots \wedge P(t_n)) \Rightarrow P(f(t_1, \dots, t_n))}{\forall t \in T_\Sigma. P(t)}$$

Structural Induction for Commands



Given the syntax of commands:

$$c \in Com ::= \text{skip} \mid x := a \mid c; c \mid \text{if } b \text{ then } c \text{ else } c \mid \text{while } b \text{ do } c$$

We have that structural induction is as follows:

$$\frac{\begin{array}{l} P(\text{skip}) \quad \forall x, a. P(x := a) \\ \forall c_0, c_1. P(c_0) \wedge P(c_1) \Rightarrow P(c_0; c_1) \\ \forall b, c_0, c_1. P(c_0) \wedge P(c_1) \Rightarrow P(\text{if } b \text{ then } c_0 \text{ else } c_1) \\ \forall b, c. P(c) \Rightarrow P(\text{while } b \text{ do } c) \end{array}}{\forall c \in Com. P(c)}$$

Determinacy by Structural Induction



Base Cases

$P(\text{skip})$

$\forall x, a. P(x := a)$

Inductive Cases

$\forall c_0, c_1. P(c_0) \wedge P(c_1) \Rightarrow P(c_0; c_1)$

$\forall b, c_0, c_1. P(c_0) \wedge P(c_1) \Rightarrow P(\text{if } b \text{ then } c_0 \text{ else } c_1)$

$\forall b, c. P(c) \Rightarrow P(\text{while } b \text{ do } c)$

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$\forall b, c_0, c_1. P(c_0) \wedge P(c_1) \Rightarrow P(\text{if } b \text{ then } c_0 \text{ else } c_1)$

$\forall b, c. P(c) \Rightarrow P(\text{while } b \text{ do } c)$

The case for **while** b **do** c fails, due to the recursive definition of its semantics:

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$

one of the premises is as complex as the conclusion!

Where is the Problem?



$$\forall b, c. P(c) \Rightarrow P(\text{while } b \text{ do } c)$$

- Consider arbitrary b and c . Our inductive hypothesis:
 $P(c) \triangleq \forall \sigma, \sigma_1, \sigma_2 \in \mathbb{M}. (\langle c, \sigma \rangle \longrightarrow \sigma_1 \wedge \langle c, \sigma \rangle \longrightarrow \sigma_2) \Rightarrow \sigma_1 = \sigma_2$

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- We want to prove

$$P(\text{while } b \text{ do } c) \triangleq \forall \sigma, \sigma_1, \sigma_2 \in \mathbb{M}.$$

$$(\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1 \wedge \langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2) \Rightarrow \sigma_1 = \sigma_2$$

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- ▶ Take $\sigma, \sigma_1, \sigma_2$ such that $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1$ and $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
We want to prove $\sigma_1 = \sigma_2$.

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- ▶ Take $\sigma, \sigma_1, \sigma_2$ such that $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1$ and $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
We want to prove $\sigma_1 = \sigma_2$.
- ▶ By **determinacy of boolean expressions**, there are two cases:
 $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and $\langle b, \sigma \rangle \longrightarrow \text{ff}$. The issue is when $\langle b, \sigma \rangle \longrightarrow \text{tt}$.

Where is the Problem? (cont.)



- Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1$, assuming $\langle b, \sigma \rangle \longrightarrow \text{tt}$.

Where is the Problem? (cont.)



- ▶ Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1$, assuming $\langle b, \sigma \rangle \longrightarrow \text{tt}$.
- ▶ The only applicable rule is

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$

hence $\sigma_1 = \sigma'_1$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_1$ and $\langle \text{while } b \text{ do } c, \sigma''_1 \rangle \longrightarrow \sigma'_1$.

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hence $\sigma_1 = \sigma'_1$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_1$ and $\langle \text{while } b \text{ do } c, \sigma''_1 \rangle \longrightarrow \sigma'_1$.

- ▶ Similarly, since $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$,
it must be $\sigma_2 = \sigma'_2$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_2$ and $\langle \text{while } b \text{ do } c, \sigma''_2 \rangle \longrightarrow \sigma'_2$.

Where is the Problem? (cont.)



► Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1$, assuming $\langle b, \sigma \rangle \longrightarrow \text{tt}$.

► The only applicable rule is

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$

hence $\sigma_1 = \sigma'_1$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_1$ and $\langle \text{while } b \text{ do } c, \sigma''_1 \rangle \longrightarrow \sigma'_1$.

► Similarly, since $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$,

it must be $\sigma_2 = \sigma'_2$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_2$ and $\langle \text{while } b \text{ do } c, \sigma''_2 \rangle \longrightarrow \sigma'_2$.

► By the inductive hypothesis (IH) $P(c)$, we have $\sigma''_1 = \sigma''_2$.

Where is the Problem? (cont.)



- ▶ Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_1$, assuming $\langle b, \sigma \rangle \longrightarrow \text{tt}$.
- ▶ The only applicable rule is

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$

hence $\sigma_1 = \sigma'_1$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_1$ and $\langle \text{while } b \text{ do } c, \sigma''_1 \rangle \longrightarrow \sigma'_1$.

- ▶ Similarly, since $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$,
it must be $\sigma_2 = \sigma'_2$ with $\langle c, \sigma \rangle \longrightarrow \sigma''_2$ and $\langle \text{while } b \text{ do } c, \sigma''_2 \rangle \longrightarrow \sigma'_2$.
- ▶ By the inductive hypothesis (IH) $P(c)$, we have $\sigma''_1 = \sigma''_2$.
- ▶ Thus, $\langle \text{while } b \text{ do } c, \sigma''_2 \rangle \longrightarrow \sigma'_1$ and $\langle \text{while } b \text{ do } c, \sigma''_2 \rangle \longrightarrow \sigma'_2$, but
there is no inductive hypothesis $P(\text{while } b \text{ do } c)$!

A Recursive Definition!



this premise is as complex as the conclusion!

$$\frac{\langle b, \sigma \rangle \longrightarrow \texttt{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \texttt{while } b \texttt{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \texttt{while } b \texttt{ do } c, \sigma \rangle \longrightarrow \sigma'}$$

To prove determinacy for commands we need another induction principle:
rule induction.

Derivations

A **logical system** is a set of axioms and inference rules:

$$R = \left\{ \frac{\quad}{z}, \frac{x_1 \quad \cdots \quad x_n}{y}, \dots \right\}$$

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A **derivation** in the logical system R is written $d \Vdash_R y$ where either

- ▶ $d = \left(\frac{}{y} \right)$ is an axiom of R ;
- ▶ or $d = \left(\frac{d_1 \cdots d_n}{y} \right)$ for some derivations $d_1 \Vdash_R x_1, \dots, d_n \Vdash_R x_n$
such that $\left(\frac{x_1 \cdots x_n}{y} \right)$ is an inference rule of R .

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We define $D_R \triangleq \{d \mid d \Vdash_R y\}$.

Immediate Subderivation Relation



$$A = D_R$$

$$\prec = \left\{ \left(d_i, \frac{d_1 \cdots d_n}{y} \right) \mid d_1 \Vdash_R x_1, \dots, d_n \Vdash_R x_n, \left(\frac{x_1 \cdots x_n}{y} \right) \in R \right\}$$

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Example

$$R = \left\{ \frac{}{N \longrightarrow n}, \frac{E_0 \longrightarrow n_0 \quad E_1 \longrightarrow n_1}{E_0 \oplus E_1 \longrightarrow n_0 + n_1}, \frac{E_0 \longrightarrow n_0 \quad E_1 \longrightarrow n_1}{E_0 \otimes E_1 \longrightarrow n_0 \cdot n_1} \right\}$$

$$\frac{}{2 \longrightarrow 2} \prec \frac{\frac{}{1 \longrightarrow 1} \quad \frac{}{2 \longrightarrow 2}}{(1 \oplus 2) \longrightarrow 3} \prec \frac{\frac{\frac{}{1 \longrightarrow 1} \quad \frac{}{2 \longrightarrow 2}}{(1 \oplus 2) \longrightarrow 3} \quad \frac{\frac{}{3 \longrightarrow 3} \quad \frac{}{4 \longrightarrow 4}}{(3 \oplus 4) \longrightarrow 7}}{(1 \oplus 2) \otimes (3 \oplus 4) \longrightarrow 21}$$

Measuring Derivations



Let $\text{height} : D_R \rightarrow \mathbb{N}$ be defined as:

$$\text{height}\left(\frac{\quad}{y}\right) \triangleq 1 \quad \text{if } \left(\frac{\quad}{y}\right) \in R$$

$$\text{height}\left(\frac{d_1, \dots, d_n}{y}\right) \triangleq 1 + \max_{i \in [1, n]} \text{height}(d_i) \quad \text{if } d_1 \Vdash_R x_1, \dots, d_n \Vdash_R x_n, \left(\frac{x_1 \cdots x_n}{y}\right) \in R$$

Example

$$\text{height}\left(\frac{\quad}{2 \longrightarrow 2}\right) = 1 \quad \text{height}\left(\frac{\frac{1 \longrightarrow 1 \quad 2 \longrightarrow 2}{(1 \oplus 2) \longrightarrow 3}}{\quad}\right) = 2$$

$\prec$ on Derivations is Well-Founded



- ▶ The measure `height` is useful to connect $\prec$ with well-founded relations for $\mathbb{N}$.
- ▶ By definition, if $d \prec d'$ then `height`(d) $<$ `height`(d').
- ▶ Any descending chain in $\prec$ induces a descending chain in $<$.
- ▶ Since $<$ is well-founded so is $\prec$.

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- ▶ The measure **height** is useful to connect $\prec$ with well-founded relations for $\mathbb{N}$.
- ▶ By definition, if $d \prec d'$ then $\text{height}(d) < \text{height}(d')$.
- ▶ Any descending chain in $\prec$ induces a descending chain in $<$.
- ▶ Since $<$ is well-founded so is $\prec$.

Consider $\prec^+$, the **transitive closure** of $\prec$. We have, e.g.,

$$\overline{2 \longrightarrow 2} \prec^+ \frac{\overline{1 \longrightarrow 1} \quad \overline{2 \longrightarrow 2} \quad \overline{3 \longrightarrow 3} \quad \overline{4 \longrightarrow 4}}{\overline{(1 \oplus 2) \longrightarrow 3} \quad \overline{(3 \oplus 4) \longrightarrow 7}} \frac{}{\overline{(1 \oplus 2) \otimes (3 \oplus 4) \longrightarrow 21}}$$

- ▶ **Corollary:** $\prec^+$ is well-founded.



Because $\prec$ is well-founded, we can now instantiate the induction principle!

$$\frac{\forall \left(\frac{x_1 \cdots x_n}{y} \right) \in R. \forall d_i \Vdash_R x_i. (P(d_1) \wedge \cdots \wedge P(d_n)) \Rightarrow P \left(\frac{d_1 \cdots d_n}{y} \right)}{\forall d. P(d)}$$

A Variant: Rule Induction



Recall: $I_R \triangleq \{y \mid \Vdash_R y\}$ is the set of all theorems of R .

$$\frac{\forall \left(\frac{x_1 \cdots x_n}{y} \right) \in R. (\{x_1, \dots, x_n\} \subseteq I_R \wedge P(x_1) \wedge \cdots \wedge P(x_n)) \Rightarrow P(y)}{\forall x \in I_R. P(x)}$$

A Variant: Rule Induction



Recall: $I_R \triangleq \{y \mid \Vdash_R y\}$ is the set of all theorems of R .

$$\frac{\forall \left(\frac{x_1 \cdots x_n}{y} \right) \in R. (\{x_1, \dots, x_n\} \subseteq I_R \wedge P(x_1) \wedge \cdots \wedge P(x_n)) \Rightarrow P(y)}{\forall x \in I_R. P(x)}$$

Having $\{x_1, \dots, x_n\} \subseteq I_R$ means assuming a derivation d_i for each theorem x_i .
Without this assumption, we have a simplified variant:

$$\frac{\forall \left(\frac{x_1 \cdots x_n}{y} \right) \in R. (P(x_1) \wedge \cdots \wedge P(x_n)) \Rightarrow P(y)}{\forall x \in I_R. P(x)}$$



Properties of numbers $P(n) \rightsquigarrow$ **Mathematical induction**

Two proof obligations: $P(0)$ and $P(n) \Rightarrow P(n + 1)$



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One proof obligation for each function symbol



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Properties of terms $P(t) \rightsquigarrow$ **Structural induction**

One proof obligation for each function symbol

Properties of formulas $P(F) \rightsquigarrow$ **Rule induction**

One proof obligation for each inference rule

Two Views of Determinacy



Properties of terms $P(t) \rightsquigarrow$ **Structural induction**

$$P(c) \triangleq \forall \sigma, \sigma_1, \sigma_2 \in \mathbb{M}. (\langle c, \sigma \rangle \longrightarrow \sigma_1 \wedge \langle c, \sigma \rangle \longrightarrow \sigma_2) \Rightarrow \sigma_1 = \sigma_2$$

Two Views of Determinacy



Properties of terms $P(t) \rightsquigarrow$ **Structural induction**

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Properties of formulas $P(F) \rightsquigarrow$ **Rule induction**

$$P(\langle c, \sigma \rangle \longrightarrow \sigma_1) \triangleq \forall \sigma_2 \in \mathbb{M}. \langle c, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma_1 = \sigma_2$$

IMP Semantics (Commands)



$$\begin{array}{c} \frac{}{\langle \text{skip}, \sigma \rangle \longrightarrow \sigma} \quad \frac{\langle a, \sigma \rangle \longrightarrow n}{\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]} \quad \frac{\langle c_0, \sigma \rangle \longrightarrow \sigma'' \quad \langle c_1, \sigma'' \rangle \longrightarrow \sigma'}{\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma'} \\[10pt] \frac{\langle b, \sigma \rangle \longrightarrow \text{ff} \quad \langle c_1, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'} \quad \frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c_0, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'} \\[10pt] \frac{\langle b, \sigma \rangle \longrightarrow \text{ff}}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma} \quad \frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'} \end{array}$$

- $P(\langle c, \sigma \rangle \longrightarrow \sigma_1) \triangleq \forall \sigma_2 \in \mathbb{M}. \langle c, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma_1 = \sigma_2$
- $\forall c, \sigma, \sigma_1. P(\langle c, \sigma \rangle \longrightarrow \sigma_1)?$

Determinacy: Base Case #1



$$\overline{\langle \text{skip}, \sigma \rangle \longrightarrow \sigma}$$

We want to prove

$$P(\langle \text{skip}, \sigma \rangle \longrightarrow \sigma) \triangleq \forall \sigma_2. \langle \text{skip}, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma = \sigma_2$$

Take σ_2 such that $\langle \text{skip}, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma = \sigma_2$.

Determinacy: Base Case #1



$$\overline{\langle \text{skip}, \sigma \rangle \longrightarrow \sigma}$$

We want to prove

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Take σ_2 such that $\langle \text{skip}, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma = \sigma_2$.

- ▶ Consider the goal $\langle \text{skip}, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ Only the rule $\overline{\langle \text{skip}, \sigma \rangle \longrightarrow \sigma}$ is applicable, hence $\sigma = \sigma_2$.

Determinacy: Base Case #2



$$\frac{\langle a, \sigma \rangle \longrightarrow n}{\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]}$$

We assume $\langle a, \sigma \rangle \longrightarrow n$. We want to prove

$$P(\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]) \triangleq \forall \sigma_2. \langle x := a, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma[n/x] = \sigma_2$$

Take σ_2 such that $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma[n/x] = \sigma_2$.

Determinacy: Base Case #2

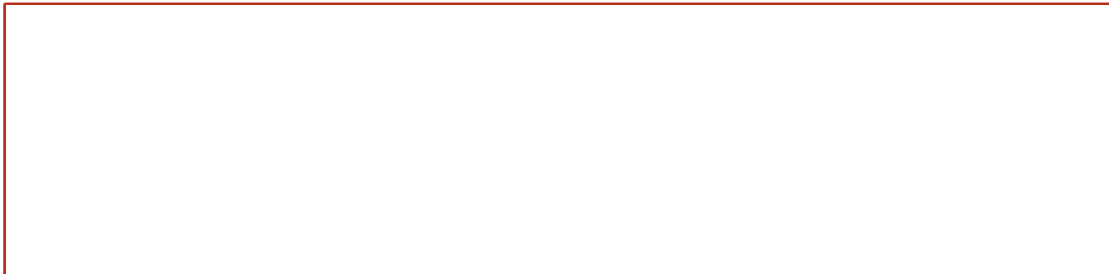


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Take σ_2 such that $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma[n/x] = \sigma_2$.



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Take σ_2 such that $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma[n/x] = \sigma_2$.

► Consider the goal $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$.

Determinacy: Base Case #2



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Take σ_2 such that $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma[n/x] = \sigma_2$.

- ▶ Consider the goal $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ Only the rule $\frac{\langle a, \sigma \rangle \longrightarrow n}{\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]}$ is applicable, hence $\sigma_2 = \sigma[n/x]$, with $\langle a, \sigma \rangle \longrightarrow n$.

Determinacy: Base Case #2



$$\frac{\langle a, \sigma \rangle \longrightarrow n}{\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]}$$

We assume $\langle a, \sigma \rangle \longrightarrow n$. We want to prove

$$P(\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]) \triangleq \forall \sigma_2. \langle x := a, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma[n/x] = \sigma_2$$

Take σ_2 such that $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma[n/x] = \sigma_2$.

- ▶ Consider the goal $\langle x := a, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ Only the rule $\frac{\langle a, \sigma \rangle \longrightarrow n}{\langle x := a, \sigma \rangle \longrightarrow \sigma[n/x]}$ is applicable, hence $\sigma_2 = \sigma[n/x]$, with $\langle a, \sigma \rangle \longrightarrow n$.
- ▶ Since we assumed $\langle a, \sigma \rangle \longrightarrow n$, by **determinacy of arithmetic expressions** we have $n = m$, and thus $\sigma_2 = \sigma[m/x] = \sigma[n/x]$.

Determinacy: Inductive Case #1



$$\frac{\langle c_0, \sigma \rangle \longrightarrow \sigma'' \quad \langle c_1, \sigma'' \rangle \longrightarrow \sigma'}{\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma'}$$

We assume (inductive hypotheses):

$$P(\langle c_0, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle c_1, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

We want to prove $P(\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle c_0; c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2$.

Take σ_2 such that $\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma' = \sigma_2$.

Determinacy: Inductive Case #1 (cont.)



To prove: $\sigma' = \sigma_2$. We assume (inductive hypotheses):

$$P(\langle c_0, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle c_1, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

Determinacy: Inductive Case #1 (cont.)



To prove: $\sigma' = \sigma_2$. We assume (inductive hypotheses):

$$P(\langle c_0, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle c_1, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

► Consider the goal $\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma_2$.

Determinacy: Inductive Case #1 (cont.)



To prove: $\sigma' = \sigma_2$. We assume (inductive hypotheses):

$$P(\langle c_0, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle c_1, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

► Consider the goal $\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma_2$.

► Only the rule
$$\frac{\langle c_0, \sigma \rangle \longrightarrow \sigma'' \quad \langle c_1, \sigma'' \rangle \longrightarrow \sigma'}{\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma'}$$
 is applicable, hence

$$\sigma_2 = \sigma_2', \text{ with } \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \text{ and } \langle c_1, \sigma_2'' \rangle \longrightarrow \sigma_2'.$$

Determinacy: Inductive Case #1 (cont.)



To prove: $\sigma' = \sigma_2$. We assume (inductive hypotheses):

$$P(\langle c_0, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle c_1, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

- ▶ Consider the goal $\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ Only the rule
$$\frac{\langle c_0, \sigma \rangle \longrightarrow \sigma'' \quad \langle c_1, \sigma'' \rangle \longrightarrow \sigma'}{\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma'}$$
 is applicable, hence $\sigma_2 = \sigma_2'$, with $\langle c_0, \sigma \rangle \longrightarrow \sigma_2''$ and $\langle c_1, \sigma_2'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle c_0, \sigma \rangle \longrightarrow \sigma'')$, we have $\sigma'' = \sigma_2''$ and thus $\langle c_1, \sigma'' \rangle \longrightarrow \sigma_2'$.

Determinacy: Inductive Case #1 (cont.)



To prove: $\sigma' = \sigma_2$. We assume (inductive hypotheses):

$$P(\langle c_0, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c_0, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle c_1, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

- ▶ Consider the goal $\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ Only the rule
$$\frac{\langle c_0, \sigma \rangle \longrightarrow \sigma'' \quad \langle c_1, \sigma'' \rangle \longrightarrow \sigma'}{\langle c_0; c_1, \sigma \rangle \longrightarrow \sigma'}$$
 is applicable, hence $\sigma_2 = \sigma_2'$, with $\langle c_0, \sigma \rangle \longrightarrow \sigma_2''$ and $\langle c_1, \sigma_2'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle c_0, \sigma \rangle \longrightarrow \sigma'')$, we have $\sigma'' = \sigma_2''$ and thus $\langle c_1, \sigma'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle c_1, \sigma'' \rangle \longrightarrow \sigma')$, we have $\sigma' = \sigma_2'$ and we conclude: $\sigma' = \sigma_2$.

Determinacy: Inductive Case #2



$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff} \quad \langle c_1, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'}$$

We assume $\langle b, \sigma \rangle \longrightarrow \text{ff}$ and the inductive hypothesis:

$$P(\langle c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2$$

We want to prove

$$P(\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2.$$

Take σ_2 such that $\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma' = \sigma_2$.

Determinacy: Inductive Case #2 (cont.)



We assume:

$$\langle b, \sigma \rangle \longrightarrow \text{ff}$$

$$P(\langle c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2$$

Determinacy: Inductive Case #2 (cont.)



We assume:

$$\langle b, \sigma \rangle \longrightarrow \text{ff}$$

$$P(\langle c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2$$

► Consider the goal $\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma_2$.

Determinacy: Inductive Case #2 (cont.)



We assume:

$$\langle b, \sigma \rangle \longrightarrow \text{ff}$$

$$P(\langle c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2$$

- ▶ Consider the goal $\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ By determinacy of boolean expressions, the only applicable rule is
$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff} \quad \langle c_1, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'} \text{ hence } \sigma_2 = \sigma'_2, \text{ with } \langle c_1, \sigma \rangle \longrightarrow \sigma'_2.$$

Determinacy: Inductive Case #2 (cont.)



We assume:

$$\langle b, \sigma \rangle \longrightarrow \text{ff}$$

$$P(\langle c_1, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle c_1, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2$$

- ▶ Consider the goal $\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ By determinacy of boolean expressions, the only applicable rule is
$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff} \quad \langle c_1, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'} \text{ hence } \sigma_2 = \sigma'_2, \text{ with } \langle c_1, \sigma \rangle \longrightarrow \sigma'_2.$$
- ▶ By IH $P(\langle c_1, \sigma \rangle \longrightarrow \sigma')$, we then have $\sigma' = \sigma'_2 = \sigma_2$, and we are done.

Determinacy: Inductive Case #3



$$\frac{\langle b, \sigma \rangle \longrightarrow tt \quad \langle c_0, \sigma \rangle \longrightarrow \sigma'}{\langle \text{if } b \text{ then } c_0 \text{ else } c_1, \sigma \rangle \longrightarrow \sigma'}$$

This case is analogous to the previous one.

Determinacy: Base Case #3



$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff}}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma}$$

We assume $\langle b, \sigma \rangle \longrightarrow \text{ff}$. We want to prove

$$P(\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma) \triangleq \forall \sigma_2. \langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma = \sigma_2$$

Take σ_2 such that $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma = \sigma_2$.

Determinacy: Base Case #3



$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff}}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma}$$

We assume $\langle b, \sigma \rangle \longrightarrow \text{ff}$. We want to prove

$$P(\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma) \triangleq \forall \sigma_2. \langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma = \sigma_2$$

Take σ_2 such that $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma = \sigma_2$.

- ▶ Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ By **determinacy of boolean expressions**, only the rule

$$\frac{\langle b, \sigma \rangle \longrightarrow \text{ff}}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma} \text{ is applicable, hence } \sigma_2 = \sigma.$$

Determinacy: Inductive Case #4



$$\frac{\langle b, \sigma \rangle \longrightarrow \texttt{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \texttt{while } b \texttt{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \texttt{while } b \texttt{ do } c, \sigma \rangle \longrightarrow \sigma'}$$

We assume $\langle b, \sigma \rangle \longrightarrow \texttt{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2''. \langle c, \sigma \rangle \longrightarrow \sigma_2'' \Rightarrow \sigma'' = \sigma_2''$$

$$P(\langle \texttt{while } b \texttt{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2'. \langle \texttt{while } b \texttt{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \Rightarrow \sigma' = \sigma_2'$$

We want to prove

$$P(\langle \texttt{while } b \texttt{ do } c, \sigma \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2. \langle \texttt{while } b \texttt{ do } c, \sigma \rangle \longrightarrow \sigma_2 \Rightarrow \sigma' = \sigma_2.$$

Take σ_2 such that $\langle \texttt{while } b \texttt{ do } c, \sigma \rangle \longrightarrow \sigma_2$. We want to prove that $\sigma' = \sigma_2$.

Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$

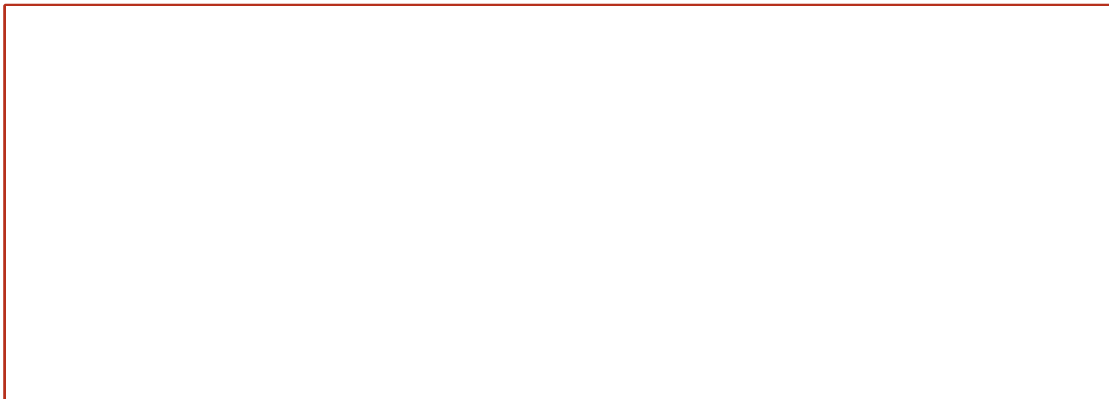
Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$



Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow tt$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$

► Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.

Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$

- Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
- By determinacy of boolean expressions, only the rule
$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$
is applicable, hence $\sigma_2 = \sigma_2'$, with $\langle c, \sigma \rangle \longrightarrow \sigma_2''$ and $\langle \text{while } b \text{ do } c, \sigma_2'' \rangle \longrightarrow \sigma_2'$.

Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$

- ▶ Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ By determinacy of boolean expressions, only the rule
$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$
is applicable, hence $\sigma_2 = \sigma_2'$, with $\langle c, \sigma \rangle \longrightarrow \sigma_2''$ and $\langle \text{while } b \text{ do } c, \sigma_2'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle c, \sigma \rangle \longrightarrow \sigma'')$, $\sigma'' = \sigma_2''$ thus $\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2'$.

Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$

- ▶ Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ By determinacy of boolean expressions, only the rule
$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$
is applicable, hence $\sigma_2 = \sigma_2'$, with $\langle c, \sigma \rangle \longrightarrow \sigma_2''$ and $\langle \text{while } b \text{ do } c, \sigma_2'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle c, \sigma \rangle \longrightarrow \sigma'')$, $\sigma'' = \sigma_2''$ thus $\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma')$, $\sigma' = \sigma_2'$ and we conclude $\sigma' = \sigma_2$.

Determinacy: Inductive Case #4 (cont.)



We assume $\langle b, \sigma \rangle \longrightarrow \text{tt}$ and the inductive hypotheses:

$$P(\langle c, \sigma \rangle \longrightarrow \sigma'') \triangleq \forall \sigma_2'' . \langle c, \sigma \rangle \longrightarrow \sigma_2'' \quad \Rightarrow \quad \sigma'' = \sigma_2''$$

$$P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma') \triangleq \forall \sigma_2' . \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2' \quad \Rightarrow \quad \sigma' = \sigma_2'$$

- ▶ Consider the goal $\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma_2$.
- ▶ By determinacy of boolean expressions, only the rule
$$\frac{\langle b, \sigma \rangle \longrightarrow \text{tt} \quad \langle c, \sigma \rangle \longrightarrow \sigma'' \quad \langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma'}{\langle \text{while } b \text{ do } c, \sigma \rangle \longrightarrow \sigma'}$$
is applicable, hence $\sigma_2 = \sigma_2'$, with $\langle c, \sigma \rangle \longrightarrow \sigma_2''$ and $\langle \text{while } b \text{ do } c, \sigma_2'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle c, \sigma \rangle \longrightarrow \sigma'')$, $\sigma'' = \sigma_2''$ thus $\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma_2'$.
- ▶ By IH $P(\langle \text{while } b \text{ do } c, \sigma'' \rangle \longrightarrow \sigma')$, $\sigma' = \sigma_2'$ and we conclude $\sigma' = \sigma_2$.
- ▶ This concludes the case (and the proof of determinacy).



The End